

Table 1: **Vina score** of Top-100 diverse modes on LIT-PCBA. Mean $\pm$ standard deviations over 3 runs are reported.

Method		GBA	IDH1	KAT2A	MAPK1	MTORC1
Reaction	SynFlowNet [†]	-9.27 $\pm$ 0.06	-10.40 $\pm$ 0.08	-9.41 $\pm$ 0.04	-8.92 $\pm$ 0.05	-10.84 $\pm$ 0.03
	RGFN [†]	-8.48 $\pm$ 0.06	-9.49 $\pm$ 0.13	-8.53 $\pm$ 0.11	-8.22 $\pm$ 0.15	-9.89 $\pm$ 0.06
	RxnFlow [†]	-9.62 $\pm$ 0.04	-10.95 $\pm$ 0.05	-9.73 $\pm$ 0.03	-9.30 $\pm$ 0.01	-11.39 $\pm$ 0.09
SMILES	S3-GFN	-9.76 $\pm$ 0.08	-11.43 $\pm$ 0.03	-10.02 $\pm$ 0.10	-9.38 $\pm$ 0.03	-11.73 $\pm$ 0.01
Method		OPRK1	PKM2	PPARG	TP53	VDR
Reaction	SynFlowNet [†]	-10.34 $\pm$ 0.07	-11.98 $\pm$ 0.12	-9.40 $\pm$ 0.05	-7.90 $\pm$ 0.10	-11.62 $\pm$ 0.13
	RGFN [†]	-9.61 $\pm$ 0.11	-10.96 $\pm$ 0.18	-8.53 $\pm$ 0.07	-7.07 $\pm$ 0.06	-10.86 $\pm$ 0.11
	RxnFlow [†]	-10.84 $\pm$ 0.03	-12.53 $\pm$ 0.02	-9.73 $\pm$ 0.02	-8.09 $\pm$ 0.06	-12.30 $\pm$ 0.07
SMILES	S3-GFN	-11.48 $\pm$ 0.01	-13.15 $\pm$ 0.03	-9.89 $\pm$ 0.02	-8.33 $\pm$ 0.10	-12.86 $\pm$ 0.10

Table 2: **Results on sEH + QED**. Mean and standard deviations are reported with three independent runs.

	Positive ($\uparrow$)	Pos. Top100 sEH ($\uparrow$)	Pos. Top100 QED ($\uparrow$)	Pos. Top100 Div. ($\uparrow$)
Prior	0.299 $\pm$ 0.014	0.628 $\pm$ 0.008	0.482 $\pm$ 0.003	0.877 $\pm$ 0.001
S3-GFN (sEH)	0.945 $\pm$ 0.009	1.043 $\pm$ 0.001	0.664 $\pm$ 0.018	0.764 $\pm$ 0.000
S3-GFN (sEH + QED)	0.991 $\pm$ 0.002	1.001 $\pm$ 0.002	0.937 $\pm$ 0.002	0.759 $\pm$ 0.003

Table 3: **Performance under harsher constraint changes**. Mean $\pm$ standard deviation over 3 runs are reported.

	$ \mathcal{R} = 15$ # positives in buffer: 2241			$ \mathcal{R} = 10$ # positives in buffer: 1073		
	Zero-shot	RTB + RS	S3-GFN	Zero-shot	RTB + RS	S3-GFN
Avg. sEH ($\uparrow$)	1.010 $\pm$ 0.001	0.943 $\pm$ 0.001	1.007 $\pm$ 0.001	1.010 $\pm$ 0.001	0.862 $\pm$ 0.042	1.007 $\pm$ 0.002
Positive Ratio ($\uparrow$)	0.620 $\pm$ 0.023	0.865 $\pm$ 0.020	0.838 $\pm$ 0.012	0.174 $\pm$ 0.018	0.637 $\pm$ 0.061	0.844 $\pm$ 0.010
Num. Unique ($\uparrow$)	923.7 $\pm$ 3.1	795.0 $\pm$ 8.5	861.3 $\pm$ 13.7	923.7 $\pm$ 3.1	550.7 $\pm$ 22.4	687.3 $\pm$ 14.4
Pos. Top100 sEH ($\uparrow$)	1.035 $\pm$ 0.002	1.029 $\pm$ 0.001	1.037 $\pm$ 0.000	1.035 $\pm$ 0.002	1.016 $\pm$ 0.004	1.038 $\pm$ 0.000

Table 4: **Comparison on sEH under Enamine REAL reactions**. Mean $\pm$ standard deviation over 3 runs are reported.

Method	Positive ($\uparrow$)	Pos. Top100 sEH ($\uparrow$)	sEH ($\uparrow$)	Diversity ($\uparrow$)	SA ($\downarrow$)	AiZynth ($\uparrow$)	QED ($\uparrow$)	Mol. weight ($\downarrow$)
SFN (hb)	1.0	0.989 $\pm$ 0.005	0.907 $\pm$ 0.007	0.801 $\pm$ 0.022	2.876 $\pm$ 0.150	0.727 $\pm$ 0.255	0.694 $\pm$ 0.054	356.00 $\pm$ 6.31
S3-GFN (hb)	0.945 $\pm$ 0.009	1.043 $\pm$ 0.001	1.009 $\pm$ 0.000	0.764 $\pm$ 0.000	2.364 $\pm$ 0.016	0.990 $\pm$ 0.008	0.696 $\pm$ 0.007	350.35 $\pm$ 2.84
S3-GFN (REAL)	0.953 $\pm$ 0.009	1.054 $\pm$ 0.001	1.023 $\pm$ 0.002	0.757 $\pm$ 0.004	2.405 $\pm$ 0.013	0.980 $\pm$ 0.014	0.630 $\pm$ 0.002	372.99 $\pm$ 0.87